

SAI PRANAY TADAKAMALLA

B.Tech, Artificial Intelligence & Machine Learning | Founder & CEO, 221B Labs Private Limited
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ABOUT

Final-year AI & ML undergraduate at Malla Reddy University and full-stack engineer, with three papers on SSRN spanning agentic AI, cooperative multi-agent robotics and software testing. Founder and CEO of 221B Labs Private Limited, which runs campus event ticketing and is building a speech-based reading platform for Indian classrooms. Interests: reinforcement learning, agentic AI, quantum-inspired algorithms and speech technology for Indian languages.

EDUCATION

B.Tech, Artificial Intelligence & Machine Learning — Malla Reddy University, Hyderabad | CGPA 8.85 / 10 2023 – 2027 (exp.)
Class XII — Geetanjali Junior College, Nalgonda | 86.4% 2023
Class X — Kendriya Vidyalaya, Nalgonda | 76.8% 2021

RESEARCH EXPERIENCE

Undergraduate Researcher — Dept. of AI & ML, Malla Reddy University | Guide: Manasa Chandupatla 2026 – Present

- Major project: *Tunnelling Through Deception*, a quantum-inspired exploration plug-in for escaping deceptive reward traps; in a 40-seed pre-registered study, classical and quantum variants scored 0.875 and 0.877 against a 0.494 baseline ($p < 0.0001$).
- Built a reproducible, tested Python codebase and live comparative demo (github.com/PranayTadakamalla/final-project); manuscript in preparation for IEEE submission.

Student Researcher — School of Engineering, Malla Reddy University | AI for Cybersecurity 2024 – 2025

- First author of *Cyber Chat Bot using GPT Model*, on a GPT-3.5 security advisor (MERN stack, [GitHub](https://github.com)) with authentication and NLP security filters, tested with real users.
- Built *Quantum Messenger*, a quantum key-exchange and encryption simulation, and a BERT + LSTM intrusion-detection pipeline evaluated on 1,000+ simulated security events.

PUBLICATIONS (PREPRINTS)

- Tadakamalla, S. P., & Terala, S. (2026).** *The Fullness That Remains: Quantum Non-Locality, Mathematical Infinity, and the Search for an Undivided Whole*. SSRN. doi:10.2139/ssrn.7350098
- Tadakamalla, S. P., & Edukoju, G. S. (2025).** *From Pixels to Paths: Cooperative AI for Autonomous Agricultural Navigation*. SSRN. doi:10.2139/ssrn.5935575
- Chandupatla, M., **Tadakamalla, S. P.**, Prakash, S. M. R., & Y, D. (2025). *A Unified Framework for End-to-End Testing in E-Commerce Applications: Towards AI-Driven Resilience and Automation*. SSRN. doi:10.2139/ssrn.5602510

EXPERIENCE

Founder & CEO — 221B Labs Private Limited, Hyderabad | Incubated at Malla Reddy University Mar 2026 – Present

- Lead strategy, product, client delivery and hiring; run an engineering internship programme.
- Built Festora, the company's event ticketing platform, used to ticket 5,000+ students; official ticketing and community partner of Malla Reddy University and DataForge, a Microsoft developer community in Hyderabad.
- Delivered MALCON '26 registration and QR check-in for 700+ participants, covering entry and two-day meal passes.

Project Manager Intern — Pramila Foundation Jan 2025 – Jun 2025

- Led development of the organisation's website and mobile app; feature and UX changes raised user engagement by 85%.

Web Developer — Unifesto, student-led event startup | built its event registration, scheduling and login system 2025

PROJECTS

Pathana Sakthi — 221B Labs Private Limited | Speech AI, TTS, G2P, Android 2026 – Present

- Offline read-along tutor for Classes 1–10 in Telugu, English and Hindi, aligned to the SCERT curriculum, with teacher, parent and admin dashboards.
- Developing in-house kid-calibrated Indian voices, a language model and grapheme-to-phoneme (G2P) conversion; pronunciation is scored word by word through CTC forced alignment.
- Pilot plan with SCERT: 2–3 government schools, 150–300 students, targeting 75%+ read-along accuracy and 80%+ weekly teacher use.

Bazinga Labs — LLM Code Generation Platform | React.js, Node.js, LLMs | [GitHub](https://github.com)

- Generates code in Python, Java, TypeScript, React, AngularJS and Node.js through a secure, scalable API backend.

AADHARVA — AI Digital Hub for Rural Advancement | Next.js, Python, Google APIs | [GitHub](https://github.com)

- Multilingual conversational assistant that guides rural users through digital services.

ACHIEVEMENTS & LEADERSHIP

2nd Place, Pitch Arena - Ideathon 4.0 — Malla Reddy University Aug 2026

- Pitched Pathana Sakthi; advanced to Round 2 of the National Entrepreneurship Challenge 2026 (E-Cell, IIT Bombay).

Vice-President — Microsoft Learn Student Chapter, Malla Reddy University Apr 2025 – Oct 2025

- Led a 20-member team that ran 10+ workshops on AI, cloud and software development, reaching 5,000+ students.

Member — Indian Knowledge Systems (IKS) Club, Malla Reddy University 2024 – Present

SKILLS, CERTIFICATIONS & LANGUAGES

AI/ML: Machine Learning, Reinforcement Learning, NLP, LLMs, Prompt Engineering, Speech Recognition, Experimental Design
Programming & Web: Python, Java, C/C++, JavaScript, TypeScript, React.js, Next.js, Node.js, Express.js, Django, Spring Boot, REST APIs
Data, Cloud & Tools: MongoDB, MySQL, AWS, Google Cloud Platform, CI/CD, Git, [GitHub](https://github.com), Postman, Salesforce CRM
Certifications: Elements of AI (Univ. of Helsinki), Reinforcement Learning (NPTEL), Generative AI (Microsoft & LinkedIn), Responsible AI (Google), Managing Projects (PMI), Project Management (Microsoft)
Languages: English (professional, Cambridge B2), Hindi (full professional, C1), Telugu (native)